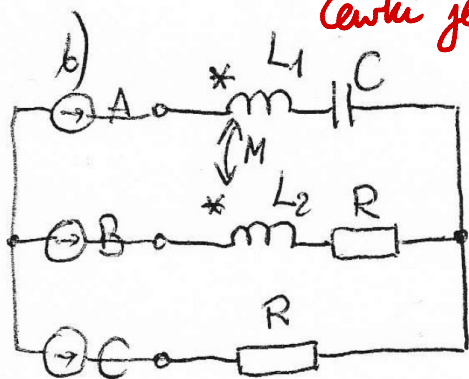


ĆWICZENIA 11

Cewki jednocześnie



$$|E_f| = 200V$$

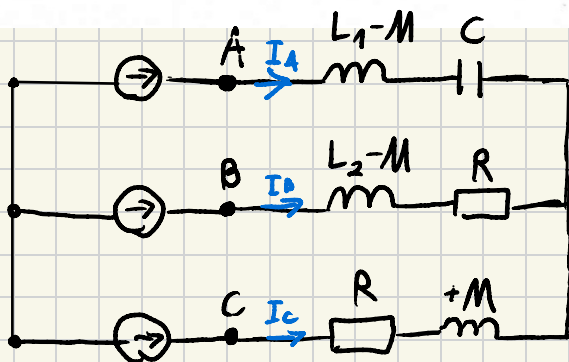
$$R = 10\Omega$$

$$X_{L1} = 30\Omega$$

$$X_{L2} = 20\Omega$$

$$X_M = 10\Omega$$

$$X_C = 20\Omega$$



$$E_A = 200 [V]$$

$$E_B = 200\left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) = -100 - j100\sqrt{3}$$

$$E_C = 200\left(-\frac{1}{2} + j\frac{\sqrt{3}}{2}\right) = -100 + j100\sqrt{3}$$

$$Z_{L1M} = Z_{L1} - Z_M = j30 - j10 = j20 [\Omega]$$

$$Z_{L2M} = Z_{L2} - Z_M = j20 - j10 = j10 [\Omega]$$

$$Z_C = R + Z_M = 10 + j10 [\Omega]$$

$$Z_B = Z_{L2M} + R = 10 + j10 [\Omega]$$

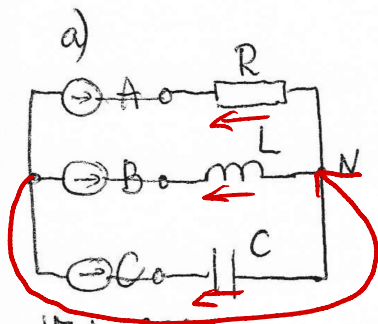
$$Z_A = Z_{L1M} + Z_C = j20 - j20 = 0$$

$$\begin{aligned}
 I_B &= (E_B - U_N) \cdot Y_B = (-100 - j100\sqrt{3} - 200) \cdot \frac{1}{10 + j10} = \\
 &= (-300 - j100\sqrt{3})(0,05 - j0,05) = -15 + j15 - j5\sqrt{3} - 5\sqrt{3} = \\
 &= -15 - 5\sqrt{3} + j(15 - 5\sqrt{3})
 \end{aligned}$$

$$\begin{aligned}
 I_C &= (E_C - U_N) \cdot Y_C = (-100 + j100\sqrt{3} - 200) \cdot \frac{1}{10 + j10} = \\
 &= (-300 + j100\sqrt{3})(0,05 - j0,05) = \\
 &= -15 + j15 + j5\sqrt{3} + 5\sqrt{3} = -15 + 5\sqrt{3} + j(15 + 5\sqrt{3})
 \end{aligned}$$

$$I_A + I_B + I_C = 0$$

$$I_A = -I_B - I_C = \dots\dots\dots$$

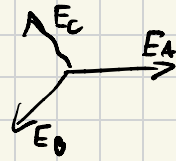


$$|E_f| = 300V$$

$$R = 10\Omega, \omega L = 10\Omega, \frac{1}{\omega C} = 10\Omega$$

$$E_f = 300V$$

$$E_A = 300V$$



$$E_B = 300\left(-\frac{1}{2} - j\frac{\sqrt{3}}{2}\right) = -150 - j150\sqrt{3}$$

$$E_C = -150 + j150\sqrt{3}$$

$$Z_A = R = 10\Omega$$

$$Z_B = Z_L = j10\Omega$$

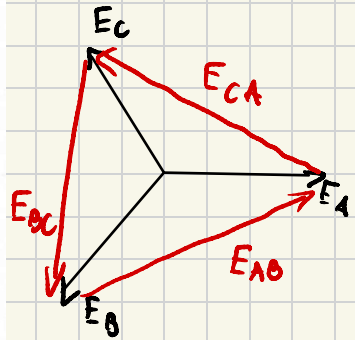
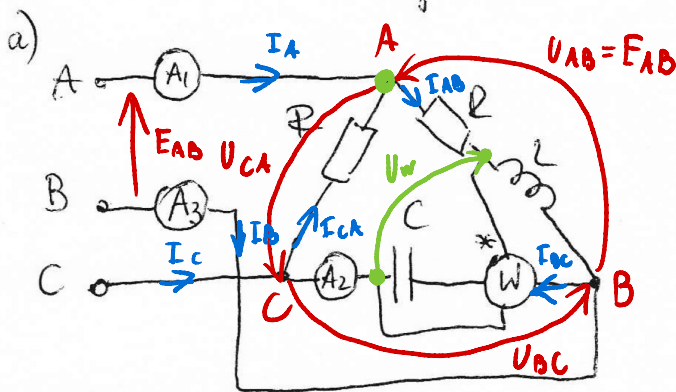
$$Z_C = Z_C = -j10\Omega$$

$$U_N = \frac{E_A \cdot Y_A + E_B \cdot Y_B + E_C \cdot Y_C}{Y_A + Y_B + Y_C} = \dots$$

$$E_A - Z_A I_A - U_N = 0 \Rightarrow I_A = (E_A - U_N) \cdot Y_A$$

$$E_B - Z_B I_B - U_N = 0 \Rightarrow I_B = \dots$$

$$E_C - Z_C I_C - U_N = 0 \Rightarrow I_C = \dots$$



$|E_f| = 400V$

$R = 10\Omega$, $X_L = 10\Omega$, $X_C = 10\Omega$

$E_{AB} = \sqrt{3} |E_f| \cdot e^{j30^\circ}$

$E_{CA} = \sqrt{3} |E_f| \cdot e^{j150^\circ}$

$E_{BC} = \sqrt{3} |E_f| \cdot e^{-j90^\circ}$

$I_A = a + jb$

$|A| = \sqrt{a^2 + b^2}$

$I_{AB} = \frac{E_{AB}}{Z_{AB}}$

$I_{CA} = \frac{E_{CA}}{Z_{CA}}$

$I_{BC} = \frac{E_{BC}}{Z_{BC}}$

$I_A - I_{CA} = I_{AB}$

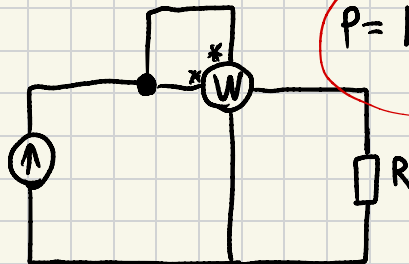
LEPSZY
TEN

RZADZIEJ
SIĘ TEGO UŻYWA

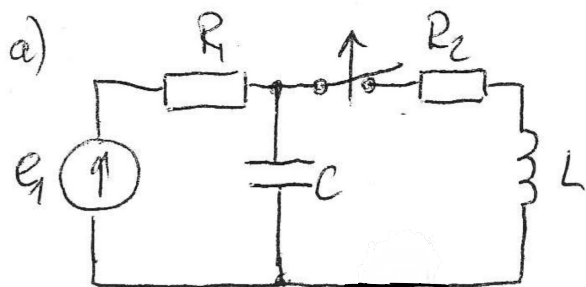
$U_W - U_L - U_C = 0$

$I_W = I_{BC}$

$P = \operatorname{Re}(U_W \cdot I_W^*)$



$P = |I|^2 \cdot R$



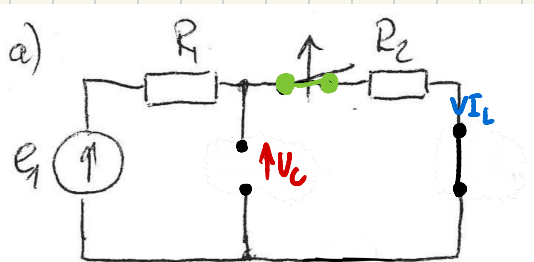
$$E_1 = 20V, \quad R_1 = 5\Omega, \quad L = 1H$$

$$R_2 = 10\Omega, \quad C = 1\mu F$$

$$u_c(t) = u_{cw}(t) + u_{cp}(t)$$

↑ ustalony
↑ przejściowy

1. Warunki początkowe



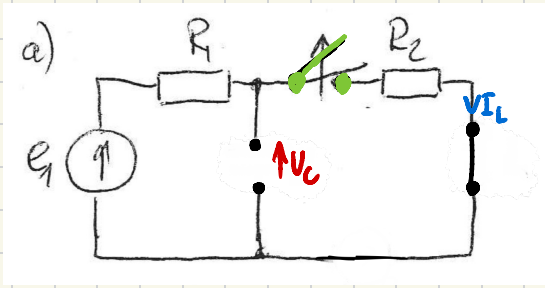
$$I_L = \frac{E_1}{R_1 + R_2} = \frac{20}{5 + 10} = \frac{20}{15} = 1,33 \text{ [A]}$$

$$u_c = R_2 \cdot I_L = 10 \cdot 1,33 = 13,33 \text{ [V]}$$

$$i_L(t) \Rightarrow i_L(0^-) = 1,33 \text{ A}$$

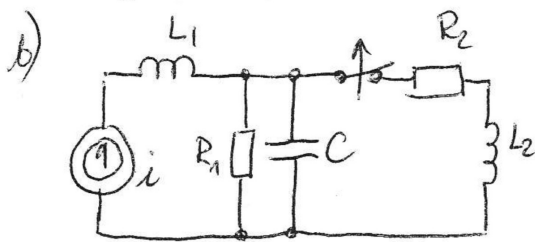
$$u_c(t) \Rightarrow u_c(0) = 13,33 \text{ V}$$

2. Stan ustalony po przełączeniu



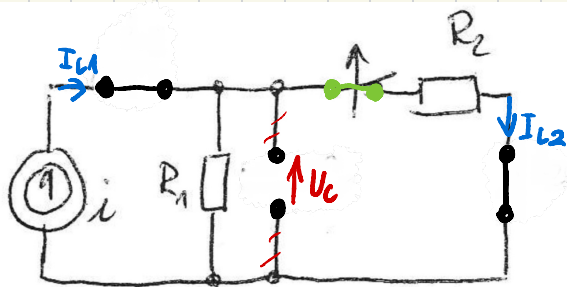
$$i_{L_u}(t) = 0 \text{ A} \Rightarrow i_{L_u}(0^+) = 0 \text{ A}$$

$$u_{C_u}(t) = e_1 = 20 \text{ V} \Rightarrow u_{C_u}(0^+) = 20 \text{ V}$$



$$\begin{aligned} i(t) &= 5A & L_1 &= 1H \\ R_1 &= 10\Omega & L_2 &= 2H \\ R_2 &= 20\Omega & C &= 10\mu F \end{aligned}$$

1. Stan początkowy PRZED PRZEŁĄCZENIEM



$$I_{L1} = 5A$$

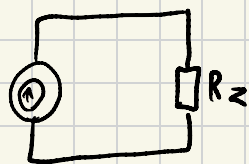
$$U_C = 33,33 V$$

$$I_{L2} = 1,667 A$$

$$i_{L1}(t) \Rightarrow i_{L1}(0^-) = 5A$$

$$i_{L2}(t) \Rightarrow i_{L2}(0^-) = 1,667A$$

$$u_C(t) \Rightarrow u_C(0^-) = 33,33 V$$



$$R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{10 \cdot 20}{10 + 20} = \frac{200}{30}$$

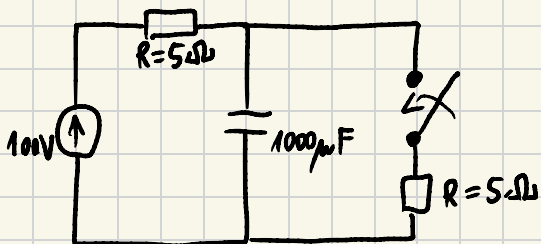
$$R_2 = 6,67 [\Omega]$$

$$U_2 = R_2 \cdot I = 6,67 \cdot 5$$

$$U_2 = 33,33 [V]$$

$$I_{L2} = \frac{U_2}{R_2} = \frac{33,33}{20} = 1,667 [A]$$

2 46 a)



$$u_C(t) = ?$$

1° War. początkowy

$$u_C(t) = 100V$$

$$u_C(0^-) = 100V$$

2° Stan ustalony po przełączeniu

$$I = \frac{E}{R_1 + R_2} = \frac{100}{10} = 10[A]$$

$$U_{R2} = I \cdot R = 5 \cdot 10 = 50[V]$$

$$u_{Cu}(t) = 50V$$

$$u_{Cu}(0^+) = 50V$$

3° Stan przejściowy

